

Day DSA Roadmap

Unit 1 – Introduction & Basics (Days 1–6)

1. Introduction to DSA, Basic Terminology, Elementary Data Organization
 2. Data Structures Operations
 3. Algorithms – Complexity & Time-Space
 4. Mathematical Notations & Functions, Algorithmic Notations
 5. Control Structures & Complexity of Algorithms
 6. Asymptotic Notations (Big O, Ω , Θ , o, ω)
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Unit 2 – Strings, Arrays, Pointers (Days 7–13)

7. String Processing – Basics, Terminology, Character Data Type
8. Strings as ADT, String Operations, Word/Text Processing
9. Pattern Matching Algorithms (Naive, KMP Intro)
10. Linear Arrays, Representation in Memory, Traversing
11. Insertion & Deletion in Arrays
12. Searching – Linear Search & Binary Search
13. Sorting – Bubble Sort, Multidimensional Arrays, Polynomial Representation

(contd.) – Records & Memory (Days 14–15)

14. Dynamic Memory Management, Records Structure, Parallel Arrays
 15. Matrices & Sparse Matrices Representation
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Unit 3 – Linked Lists, Stacks & Queues (Days 16–21)

16. Linked Lists – Introduction, Representation in Memory, Traversing & Searching
 17. Insertion & Deletion in Linked List, Header Linked List
 18. Circular Linked List & Doubly Linked List (Two-way Lists)
 19. Stacks – Introduction, Array & Linked Representation, Stack as ADT
 20. Applications of Stack – Expression Evaluation, Polish Notation, Recursion, Tower of Hanoi
 21. Queues – Linear, Linked, Circular Queue, Dequeue, Priority Queue, Applications
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Unit 4 – Trees & Graphs (Days 22–27)

22. Trees – Introduction, Binary Trees, Representations, Traversals

- 23. Binary Search Tree – Searching, Inserting, Deleting
 - 24. Balanced Binary Trees – AVL Trees
 - 25. B-Tree & B+ Tree – Searching, Insertion, Deletion
 - 26. Red-Black Trees, Heap, Heap Sort, Huffman Coding
 - 27. Graphs – Basics, Adjacency Matrix, Path Matrix, Warshall Algorithm, Traversals
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Unit 5 – Advanced Sorting, Searching & Hashing (Days 28–30)

- 28. Sorting – Insertion Sort, Selection Sort
 - 29. Sorting – Merge Sort, Shell Sort, Radix Sort
 - 30. Searching & Data Modification – Hashing Techniques + Final Revision
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